

Ex. 18. Evaluate : $\sqrt{\frac{9.5 \times .0085 \times 18.9}{.0017 \times 1.9 \times 0.021}}$.

Sol. Given exp. = $\sqrt{\frac{9.5 \times .0085 \times 18.900}{.0017 \times 1.9 \times 0.021}}$.

Now, since the sum of decimal places in the numerator and denominator under the radical sign is the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{95 \times 85 \times 18900}{17 \times 19 \times 21}} = \sqrt{5 \times 5 \times 900} = 5 \times 30 = 150.$$

Ex. 19. Simplify : $\sqrt{[(12.1)^2 - (8.1)^2] \div [(0.25)^2 + (0.25)(19.95)]}$.

(C.B.I., 2003)

Sol. Given exp. = $\sqrt{\frac{(12.1 + 8.1)(12.1 - 8.1)}{(0.25)(0.25 + 19.95)}} = \sqrt{\frac{20.2 \times 4}{0.25 \times 20.2}} = \sqrt{\frac{4}{0.25}} = \sqrt{\frac{400}{25}} = \sqrt{16} = 4.$

Ex. 20. If $x = 1 + \sqrt{2}$ and $y = 1 - \sqrt{2}$, find the value of $(x^2 + y^2)$.

Sol. $x^2 + y^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 2[(1)^2 + (\sqrt{2})^2] = 2 \times 3 = 6.$

Ex. 21. Evaluate $\sqrt{0.9}$ upto 3 places of decimal.

Sol.

$$\begin{array}{r|l} 9 & 0.900000 \text{ (0.948)} \\ & 81 \\ \hline 184 & 900 \\ & 736 \\ \hline 1888 & 16400 \\ & 15104 \\ \hline \end{array} \quad \therefore \sqrt{0.9} = 0.948.$$

Ex. 22. Find the square root of $0.\dot{1}$.

Sol. $\sqrt{0.\dot{1}} = \sqrt{\frac{1}{9}} = \frac{1}{3} = 0.333 \dots = 0.\dot{3}.$

Ex. 23. If $\sqrt{15} = 3.88$, find the value of $\sqrt{\frac{5}{3}}$.

Sol. $\sqrt{\frac{5}{3}} = \sqrt{\frac{5 \times 3}{3 \times 3}} = \frac{\sqrt{15}}{3} = \frac{3.88}{3} = 1.2933 \dots = 1.29\bar{3}.$

Ex. 24. Find the least square number which is exactly divisible by 10, 12, 15 and 18.

Sol. L.C.M. of 10, 12, 15, 18 = 180. Now, $180 = 2 \times 2 \times 3 \times 3 \times 5 = 2^2 \times 3^2 \times 5.$

To make it a perfect square, it must be multiplied by 5.

$$\therefore \text{Required number} = (2^2 \times 3^2 \times 5^2) = 900.$$

Ex. 25. Find the greatest number of five digits which is a perfect square.

Sol. Greatest number of 5 digits is 99999.

$$\begin{array}{r|l} 3 & 99999 \text{ (316)} \\ & 9 \\ \hline 61 & 99 \\ & 61 \\ \hline 626 & 3899 \\ & 3756 \\ \hline & 143 \end{array}$$

$$\therefore \text{Required number} = (99999 - 143) = 99856.$$